Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following statements is correct?

A. We can treat a bicycle wheel as a point mass when studying how the spokes flex as it rolls.
This is incorrect because spoke flex depends on the wheel's size and shape, so a point-mass model throws away the needed geometry.

B. We can reasonably treat a delivery van as a point mass when estimating how far it travels across town in 15 minutes.
For city-scale motion, the van's size is tiny compared with the travel distance, so a point-mass approximation captures its position change well.

C. We are not allowed to assume motion in a vacuum even if we want to neglect air resistance in a first-pass calculation.
This is incorrect because "ignore air resistance" is essentially the same modeling step as treating the situation like a vacuum for that effect.

D. None of the above is correct.
This is incorrect because one of the statements (about using a point-mass model over large distances) is a valid approximation.

Q2. Which of the following equations is correct?

A. 1 p = 1 £
This is incorrect because one penny is much smaller than one pound, not equal to it.

B. 1 = 0.01
This is incorrect because it equates two different numerical values without any units or conversion factor.

C. 1 p = 0.01 £
This is correct because 1 penny is defined as one hundredth of a pound.

D. 100 p = 0.01 £
This is incorrect because 100 p equals 1 £, not 0.01 £.

Q3. Which option has the incorrect unit?

A. Power: watt
This is correct because the watt (W) is the SI unit of power (J/s).

B. Pressure: newton/meter^2
This is correct because pressure is force per area, with units N/m^2 (pascal).

C. Momentum: kilogram · meter/second
This is correct because momentum is mass times velocity, with units kg · m/s.

D. Work: newton/meter
This is incorrect because work has units of N · m (joule), while N/m is the unit of a spring constant.

Q4. A scout drone is launched from a base at the origin and spots a target at the coordinates $\vec{r} = -4\hat{\imath} + 7\hat{\jmath}$, where length units are in hectometers. Which of the following statements is correct?

A. The distance between the target and the base is 11 hectometers.
This is incorrect because it adds components or guesses rather than using the Pythagorean distance formula.

B. The distance between the target and the base is 7 hectometers.
This is incorrect because it uses only the y-component and ignores the x-component.

C. The distance between the target and the base is 4 hectometers.
This is incorrect because it uses only the magnitude of the x-component and ignores the y-component.

D. None of the above is correct.
This is correct because the distance is $\sqrt{(-4)^2 + 7^2} = \sqrt{65} \approx 8.1$ hectometers, which is not 11, 7, or 4.

Q5. If the displacement of an object is given by the function $\vec{s} = \big((0.80\,\mathrm{m/s^3})t^3 + (1.9\,\mathrm{m/s^2})t^2\big)\hat{\imath}$, which of the following statements is correct?

A. This is a constant-acceleration motion.
This is incorrect because the presence of a t^3 term makes the acceleration depend on time.

B. The magnitude of the acceleration is $1.9\,\mathrm{m/s^2}$.
This is incorrect because 1.9 is a coefficient in position, not the (time-varying) acceleration.

C. The magnitude of the acceleration at $t = 0$ is $0\,\mathrm{m/s^2}$.
This is incorrect because differentiating twice leaves a nonzero constant term in acceleration.

D. The magnitude of the acceleration at $t = 0$ is $3.8\,\mathrm{m/s^2}$.
This is correct because $a(t) = \frac{d^2s}{dt^2} = (4.8t + 3.8)\,\mathrm{m/s^2}$ so at $t = 0$ it equals $3.8\,\mathrm{m/s^2}$.

Q6. An object takes 5 seconds to move from position A ($\vec{r}_A = 1.0\hat{\imath} - 3.0\hat{\jmath}$) to position B ($\vec{r}_B = -5.0\hat{\imath} + 5.0\hat{\jmath}$) along a straight path, where length units are in meters. Which of the following statements is correct?

A. The moving distance is 10 m.
This is correct because along a straight path the distance is $|\vec{r}_B - \vec{r}_A| = \sqrt{(-6)^2 + 8^2} = 10\,\mathrm{m}$.

B. The speed at $t = 0$ must be 2.0 m/s.
This is incorrect because the travel time alone only determines average speed, not the instantaneous speed at $t = 0$.

C. The acceleration at $t = 0$ must be $0.40\ \mathrm{m/s^2}$.
This is incorrect because no information about how velocity changes with time is given, so acceleration cannot be fixed.

D. None of the above is correct.
This is incorrect because the straight-line distance can be computed from the coordinates and matches one option.

Q7. A tennis ball is launched with speed u_0 at an angle α above the horizontal (ignore air resistance). What is the speed of the ball when it is at its maximum height?

A. u_0.
This is incorrect because the vertical component of velocity decreases to zero at the top, reducing the speed from its launch value.

B. $u_0 \cos \alpha$.
This is correct because at maximum height the vertical velocity is zero, so the speed equals the unchanged horizontal component $u_0 \cos \alpha$.

C. $u_0 \sin \alpha$.
This is incorrect because $u_0 \sin \alpha$ is the initial vertical component, which is not the speed at the peak.

D. 0.
This is incorrect because although the vertical velocity is zero at the top, the horizontal velocity is still nonzero.

Q8. A rescue drone is initially moving with velocity $\vec{v}_0 = (18.0\,\text{m/s})\hat{\imath}$ when its motors adjust and produce an acceleration $\vec{a} = (1.50\,\text{m/s}^2)\hat{\imath} + (3.20\,\text{m/s}^2)\hat{\jmath}$. Based on this scenario, which of the following statements is correct?

A. The x-component of the drone's velocity remains constant.
This is incorrect because a nonzero a_x changes v_x over time.

B. The y-component of the drone's velocity remains constant.
This is incorrect because a nonzero a_y changes v_y over time.

C. Both the direction of motion and speed change.
This is correct because acceleration has components in both directions, altering the velocity vector's magnitude and its direction.

D. None of the above is correct.
This is incorrect because the presence of both a_x and a_y makes one statement (about changing speed and direction) true.

Q9. If the magnitudes of both charges are tripled and their separation is doubled, what happens to the magnitude of the electrostatic force that each charge exerts on the other?

A. It stays the same.
This is incorrect because changing charges and distance changes $q_1 q_2 / r^2$ in Coulomb's law.

B. It decreases by a factor of 9/4.
This is incorrect because tripling the charges increases the force contribution from charge product rather than decreasing it.

C. It increases by a factor of 3.
This is incorrect because the net scaling is 9/4, not 3, after accounting for both charges and separation.

D. It increases by a factor of 9/4.
This is correct because $F \propto q_1 q_2 / r^2$, so $q_1 q_2$ scales by $3 \times 3 = 9$ while r^2 scales by $2^2 = 4$, giving $F' = (9/4)F$.

E. It increases by a factor of 9.
This is incorrect because it accounts for tripling the charges but forgets that doubling the separation reduces the force by a factor of 4.

Q10. An object can escape a planet with radius R and mass M at an escape speed given by $v = \sqrt{\frac{2GM}{R}}$, where G is the universal gravitational constant. Based on this expression for escape speed, which of the following statements is correct?

A. A more massive spacecraft needs a larger escape speed from the same planet.
This is incorrect because the spacecraft's mass does not appear in $v = \sqrt{2GM/R}$.

B. A planet with the same mass but a larger radius has a larger escape speed.
This is incorrect because increasing R makes $2GM/R$ smaller, lowering the escape speed.

C. A planet with smaller mass and smaller radius must always have a higher escape speed.
This is incorrect because changing M and R in opposite ways can increase or decrease M/R, so it is not "always" higher.

D. A planet with greater mass and smaller radius requires a higher escape speed.
This is correct because v increases with larger M and decreases with larger R, so larger M and smaller R both push v upward.

Q11. The force $\vec{F} = (5.50\,\text{N})\hat{\imath} + (1.20\,\text{N})\hat{\jmath}$ acts on a particle as it moves along the x axis from $x = -1.20\,\text{m}$ to $x = 2.80\,\text{m}$. What is the work, in joules, done by the force on the particle?

A. 22.0 J
This is correct because the displacement is purely in x, so $W = \vec{F} \cdot \Delta\vec{r} = F_x \Delta x = 5.50(4.00) = 22.0\,\text{J}$.

B. 15.4 J
This is incorrect because it effectively uses x_f instead of the change in position $\Delta x = x_f - x_i$.

C. 22.5 J
This is incorrect because it uses the force magnitude $|\vec{F}|$ instead of the component parallel to the motion.

D. 4.8 J
This is incorrect because it wrongly uses the y-component of force even though there is no y displacement.

Q12. A 3.2 kg cart initially moving at 1.8 m/s collides with a wall and rebounds at a speed of 0.6 m/s moving backward. The collision lasts for 0.15 seconds. What is the average force exerted on the cart during the collision?

A. 51.2 N
This is correct because $F_{\text{avg}} = \Delta p/\Delta t = m(v_f - v_i)/\Delta t = 3.2(-0.6 - 1.8)/0.15 = -51.2\,\text{N}$, so the magnitude is $51.2\,\text{N}$.

B. 38.4 N
This is incorrect because it treats the velocity change as $(-0.6) - (-1.8)$ or otherwise mishandles the sign for a rebound.

C. 64.0 N
This is incorrect because it uses an incorrect velocity change or divides by the wrong collision time.

D. 7.68 N
This is incorrect because it computes impulse $|\Delta p|$ but forgets to divide by the collision duration to get force.

Q13. A 65 kg baseball player is initially at rest. A 145 g baseball traveling at 35 m/s is caught and sticks in the glove (perfectly inelastic). What is the speed of the player immediately after the catch?

A. 0.078 m/s
This is correct because momentum conservation gives $v = \frac{m_{\text{ball}} v_0}{m_{\text{player}} + m_{\text{ball}}} = \frac{0.145 \cdot 35}{65.145} \approx 0.078\,\text{m/s}$.

B. 0.11 m/s
This is incorrect because it overestimates the result by not properly accounting for the large total mass after the catch.

C. 35 m/s
This is incorrect because it assumes the player simply takes on the ball's original speed, ignoring the player's much larger mass.

D. 2.5 m/s
This is incorrect because it confuses momentum conservation with energy conservation and produces an unrealistically large recoil speed.
